

CBSE (NCERT) vs NIOS - Class 10 Maths

Chapter comparison | kya match | kya chhoot raha | exam-important (RED)

LEGEND - kaise padhein

RED = Exam me IMPORTANT / zaroor aata hai (high weightage, must-do)

GREEN = CBSE aur NIOS DONO me COMMON hai (safe overlap - yahan strong raho)

Black = normal jaankari

EK-LINE VERDICT

Agar NCERT/CBSE achhe se padh lo to NIOS Maths ka ~85% apne aap cover.

NIOS = 26 chapters (6 module) - zyada chapters, par har chapter halka.

CBSE = 14 chapters - kam chapters, par har chapter me zyada depth + tough Q.

Sirf 2 cheez EXTRA karni hai: Commercial Maths (NIOS-only) + NIOS exam pattern/TMA.

A) CHAPTER MATCHING (CBSE chapter -> NIOS chapter)

Number System + Algebra

Real Numbers -> NIOS Ch1 Number Systems + Ch2 Exponents & Radicals

Polynomials -> NIOS Ch3 Algebraic Expr & Polynomials + Ch4 Special

Products/Factorization

Pair of Linear Eqns -> NIOS Ch5 Linear Equations

Quadratic Equations -> NIOS Ch6 Quadratic Equations

Arithmetic Progression -> NIOS Ch7 Arithmetic Progressions

Geometry

Triangles -> NIOS Ch11 Congruence + Ch14 Similarity of Triangles

Circles -> NIOS Ch15 Circles + Ch16 Angles in Circle/Cyclic Quad + Ch17

Secants/Tangents

Coordinate Geometry -> NIOS Ch19 Co-ordinate Geometry

Trigonometry

Introduction to Trig -> NIOS Ch22 Intro Trig + Ch23 Trig Ratios of Special Angles

Mensuration

Areas Related to Circles -> NIOS Ch20 Perimeters & Area of Plane Figures

Surface Areas & Volumes -> NIOS Ch21 Surface Area & Volume of Solid Figures

Statistics & Probability

Statistics -> NIOS Ch24 Data & Representation + Ch25 Measures of Central

Tendency

Probability -> NIOS Ch26 Introduction to Probability

NATIJA: CBSE ke 14 me se 13 chapters NIOS me MIL jaate hain.

B) KAUN KAUN MISS OUT HO RAHA HAI

(i) NIOS me HAI par CBSE Class 10 me NAHI (NIOS-only - ye EXTRA padho)

Ch8 Percentage & its Applications (profit-loss, discount, SI & CI) - EXAM me aata hai

Ch9 Instalment Buying (kist / EMI) - Commercial Maths, EXAM me aata hai

Ch2 Exponents & Radicals / Surds (CBSE me Class 9 level)

Ch4 Special Products & Factorization (CBSE me Class 8-9)

Ch10 Lines and Angles (CBSE Class 9)

Ch11 Congruence of Triangles (CBSE Class 9)

Ch12 Concurrent Lines (CBSE Class 9)

Ch13 Quadrilaterals (CBSE Class 9)

=> NIOS me kaafi Class-9 foundation + Commercial Maths extra juda hua hai.

(ii) CBSE Class 10 me HAI par NIOS me NAHI / bahut halka (CBSE-only)

Some Applications of Trigonometry = Heights & Distances - CBSE me hai, NIOS me dedicated chapter nahi

Real Numbers ki depth: Fundamental Theorem of Arithmetic, irrational ke proof (CBSE zyada deep)

Polynomials: zeros aur coefficients ka relation (CBSE me zyada detail)

C) VOLUME / DEPTH (kitna content)

NIOS : 26 chapters, 6 module. Zyada chapters par har chapter chhota, foundation-heavy aur questions SEEDHE (direct) hote hain.
CBSE : 14 chapters. Kam chapters par har chapter me zyada DEPTH, application aur HOTS / case-study questions.

Marks ka weightage

NIOS Maths (211): 80 marks Theory (public exam) + 20 marks TMA. Pass = 33%.

CBSE Maths: 80 marks Board theory + 20 internal. Pass = 33%.

CBSE unit-wise marks (80 ka paper) - priority ka idea

Algebra 20 marks (sabse zyada - TOP priority)

Geometry 15 marks

Trigonometry 12 marks

Statistics & Probability ... 11 marks

Mensuration 10 marks

Number Systems 06 marks

Coordinate Geometry 06 marks

D) RED = EXAM-IMPORTANT TOPICS (ye zaroor karo)

Algebra (highest weightage)

Quadratic Equations: roots, discriminant, word problems

Arithmetic Progression: nth term (a_n) aur sum (S_n)

Pair of Linear Equations: substitution/elimination + word problems

Polynomials: zeros, sum aur product of zeros

Commercial Maths (NIOs-only - na bhulna)

Simple Interest aur Compound Interest ke sums

Profit, Loss, Discount, Percentage

Instalment Buying (EMI / kist)

Geometry

Circles: tangent properties + theorems (proof + numerical)

Triangles: similarity criteria + Basic Proportionality Theorem (Thales)

Trigonometry

Trigonometric identities + ratios of 0/30/45/60/90

(CBSE) Heights & Distances ke application sums

Mensuration / Stats / Probability

Surface Area & Volume: cylinder, cone, sphere + combinations

Statistics: mean / median / mode (grouped data)

Probability: single-event basic problems

E) FINAL PLAN (NCERT base pe NIOS clear)

1. NCERT achhe se = NIOS ka ~85% done (aur strong tough base ban jaata hai)
 2. Commercial Maths (Ch8, Ch9) NIOS book se alag se padho - MUST (NCERT me nahi)
 3. Heights & Distances ek halki nazar (CBSE me hai, NIOS me optional)
 4. 3 NIOS sample paper + saari worksheets solve karo (pattern seedha hai)
 5. TMA zaroor jama karo = 20% almost-free marks
- NATIJA: NIOS Maths me sirf pass nahi - ACHHA score aayega.
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Sources: NCERT/CBSE Class 10 Maths (rationalised syllabus) + NIOS official Secondary Mathematics (211) course material (26 chapters / 6 modules).
Hinglish notes - Kiro for Krishna1k.